

Synthetic Customer V1

PREPARED BY Energy Hippo implementation team

Customer

DemoUniversityV1 represents a fictional higher-education customer with a nearby apartment complex and a central thermal plant. The geography is based around Sacramento, California so public weather station data and plausible utility territories can be used without tying the dataset to a real customer.

Working names:

- Customer: Northlake University.
- EEM companies: Northlake University, Cedar Row Apartments, Northlake Thermal Plant.
- Campus site: Northlake Main Campus.
- Housing site: Cedar Row Apartments.
- Plant site: Northlake Central Plant.
- Weather site: Weather Reference (KSAC and KSMF station meters, under Northlake University).

These names are fictional and can be changed before external demos.

Geography

- Region: Sacramento, California.
- Primary weather station: KSAC, Sacramento Executive Airport.
- Secondary weather station: KSMF, Sacramento International Airport.
- Time zone: America/Los_Angeles.
- Address coordinates are synthetic Sacramento-area values for weather and mapping setup; see [implementation/northlake-address-coordinates.pdf](#).

Utility Providers

V1 includes five fully-configured providers, deliberately split into three everyday utilities and two unique provider identities. Each provider's complete definition (services, units, channels, rate schedules, charges, file formats, environmental attributes, and scenario hooks) lives in `data/providers/<id>.yaml`, indexed by `data/providers/README.md`.

#	Synthetic provider	Class	Supplies	Data type
1	Valley Electric District	everyday	Electric	Monthly bills, 15-minute interval reads, billing demand, unmetered area lighting
2	Sierra Gas Utility	everyday	Natural gas	Monthly bills only

#	Synthetic provider	Class	Supplies	Data type
3	River City Utilities	everyday	Water + sewer + stormwater, irrigation, fire service	One bundled monthly bill with several service lines; separate irrigation and fire-service accounts
4	Northlake Thermal Plant	unique	Chilled water + steam	Internal cost allocation and hourly chilled-water interval reads
5	Helios Onsite Solar	unique	Solar generation	Generation interval reads, PPA invoice, REC / avoided-CO2e statement

The three everyday utilities cover what almost every customer has: a data-rich electric distribution utility, a deliberately simple gas utility, and a municipal biller that bundles multiple services on one account. The two unique identities exist to stress data shapes ordinary meters never produce — internal district-energy allocation, and behind-the-meter solar generation with environmental attributes.

River City Utilities consolidates what were previously two separate stubs (River City Water and Capital Sewer District) into one realistic municipal multi-service biller.

The provider names are fictional. Rate structures may be loosely based on public examples, but bills and meter data must be synthetic.

Stage 1 EEM Security Setup

The fake EEM users, companies, groups, and permission intent live in [data/security/northlake-eem-security-v1.yaml](#), with the companion note in [implementation/northlake-eem-security.pdf](#).

The DBAdmin-facing setup translation lives in [data/eem/northlake-eem-stage1-setup-v1.yaml](#). It defines the company form fields, address/contact values, the hierarchy levels every company shares (Site, Building, Meter, Point) including DBAdmin FontAwesome icon classes, deferred DBAdmin sections, and the decision to use the built-in system company `CompanyID = -1` for implementation/support users instead of creating an implementation-support company.

The follow-on MetaWorld setup translation lives in [data/eem/northlake-eem-metaworld-stage1b-v1.yaml](#). It holds only the rules that map the scenario inventory into EEM utility setup, billing account shells, meter nodes, point nodes, weather station assignments, aggregate points, and baseline shells; it never re-lists buildings, meters or points.

This is app setup data, not customer-provided bill or archive source data. It is used to prove that a blank EEMSuite seed can be configured through the actual apps before bill records, interval archives, gateway execution, workflows, or scheduled task execution are introduced.

For Stage 1, EEM companies are treated as data and security boundaries. Cedar Row assets belong to the Cedar Row company only; Northlake users who need Cedar Row visibility receive it through Cedar Row group membership rather than copied hierarchy or duplicate meters. The Northlake Thermal Plant company contains only the Central Plant; the chilled-water and steam meters at the buildings it serves belong to those university buildings, with the plant as provider.

Sites and Buildings

Site	Building	Primary use	Approximate size	Data coverage
Northlake Main Campus	Admin Hall	Office and classrooms	85,000 sq ft	Electric, gas, water + sewer + stormwater, fire service, steam
Northlake Main Campus	Science Center	Lab and classrooms	120,000 sq ft	Electric interval plus a second lab-wing service, gas, water + sewer + stormwater, fire service, chilled water + steam, solar, lab sensors
Northlake Main Campus	Library	Library and study space	70,000 sq ft	Electric (meter replacement, historian mirror), gas, water + sewer + stormwater, steam
Northlake Main Campus	Student Center	Food service and event space	95,000 sq ft	Electric interval, kitchen gas route, water + sewer + stormwater with a kitchen sewer deduct, steam, solar
Northlake Main Campus	Lakeview Residence Hall	Residence hall	110,000 sq ft, 300 beds	Electric interval, laundry submeter, gas, water + sewer + stormwater, fire service, steam
Northlake Main Campus	Recreation and Aquatics Center	Recreation and pool	65,000 sq ft	Electric interval, pool gas, water + sewer + stormwater with a pool-fill deduct, chilled water + steam
Northlake Main Campus	Parking Structure	Parking, EV charger bank	180,000 sq ft parking area	Electric, EV charger submeter
Northlake Main Campus	Campus Grounds	Outdoor service area	No floor area	Irrigation water (three meters), unmetered area lighting
Cedar Row Apartments	Cedar Row A	Multifamily housing	58,000 sq ft, 60 apartments	Electric master, water + sewer master, house gas, fire service; electric and water house submeters plus an electric and a water submeter per apartment
Cedar Row Apartments	Cedar Row B	Multifamily housing	62,000 sq ft, 64 apartments	As Cedar Row A, for 64 apartments
Cedar Row Apartments	Cedar Row Common House	Leasing, community room, laundry, pool	18,400 sq ft	Electric, gas, water + sewer with a pool-fill deduct
Cedar Row Apartments	Cedar Row Carport and Grounds	Outdoor service area	No floor area	Solar carport, unmetered parking lighting, irrigation water
Northlake Central Plant	Central Plant	District energy plant	58,400 sq ft	Electric interval (primary service), boiler gas, makeup water + sewer with a cooling-tower deduct, chilled-water, steam and condensate production, chiller controller

Water, sewer, and stormwater are billed by River City Utilities on one bundled account per building; irrigation and fire service are separate accounts, and irrigation, fire-service, Central Plant and Cedar Row accounts have no stormwater charge. Steam and chilled water are allocated by the Northlake

Thermal Plant to the university buildings it serves. Solar generation is supplied by Helios Onsite Solar and nets against Valley Electric consumption on shared buildings. Cedar Row A and Cedar Row B are master-metered: the utilities bill the property on one electric master and one water master per building, never a resident, and every apartment has an owned electric submeter and water submeter.

ENERGY STAR Property Mapping

The campus and Cedar Row sites and every building except the two grounds nodes also carry an ENERGY STAR Portfolio Manager property mapping so the dataset is ready for the EEMSuite ENERGY STAR integration. The manifest's `energyStarProperties` section (13 entries) assigns an exact PM `primaryFunction` and one or more property uses (with use details) to each of them.

Property	ENERGY STAR primary function	Property uses
Northlake Main Campus (campus parent)	College/University	College/University
Admin Hall	Office	Office + Other - Education
Science Center	Laboratory	Laboratory + Other - Education
Library	Library	Library
Student Center	Food Service	Food Service + Other - Entertainment/Public Assembly
Lakeview Residence Hall	Residence Hall/Dormitory	Residence Hall/Dormitory
Recreation and Aquatics Center	Fitness Center/Health Club/Gym	Fitness Center/Health Club/Gym + Swimming Pool
Parking Structure	Parking	Parking
Cedar Row Apartments (multifamily property)	Multifamily Housing	Multifamily Housing + Other - Recreation
Cedar Row A / B	Multifamily Housing	Multifamily Housing
Cedar Row Common House	Other - Recreation	Other - Recreation
Central Plant (standalone property)	Other - Utility	Other - Utility

The campus is modeled as a College/University parent (545,000 sq ft, parking excluded) with child building properties; Cedar Row is modeled as one multifamily property (124 units, resident population Dedicated Student) with three physical buildings: Cedar Row A, Cedar Row B and Cedar Row Common House. The Central Plant is a standalone property outside the campus boundary. See [implementation/energystar-integration.pdf](#) for the full field mapping, use-detail lists, enum values, and integration hooks.

Data Span

Initial V1 span:

- Start date: 2024-01-01.
- End date: 2025-12-31.
- Monthly bills (and the thermal plant's monthly allocation) for all provider categories.
- 15-minute electric interval reads for Science Center, Student Center, Lakeview Residence Hall, Recreation and Aquatics Center, and Central Plant.
- 15-minute solar generation interval reads for Science Center, Student Center, and the Cedar Row carport (Cedar Row Carport and Grounds).

- Hourly chilled water interval reads for Science Center and Recreation and Aquatics Center, and hourly Central Plant production (chilled water, steam, condensate).
- Hourly fixed-network submeter reads for every Cedar Row apartment and house electric submeter, the Lakeview Residence Hall laundry and the Parking Structure EV chargers.
- Monthly route reads for every Cedar Row apartment and house water submeter.

Meter And Point Inventory

data/scenarios/demo-university-v1.yaml now defines the V1 source inventory at the account, meter, and channel level. The counts below come from one join of the manifests (packet 0.3):

- 3 companies, 3 sites plus the Weather Reference site, and 13 buildings.
- 49 accounts: 40 external utility accounts, 6 internal cost centers and 3 PPA agreements.
- 322 meters: 55 utility meters (including the Library electric replacement pair), 265 owned meters (248 apartment submeters for 124 apartments plus 17 house, deduct, production, sensor, logical-source and controller meters) and 2 weather station meters.
- 431 points. Bill-only meters have no points; bill usage stays on the billing account.
- 139 related measurements (132 register/usage pairs, 4 degree-day, 3 baseline) and 7 aggregate points.
- EEMSuite gateway runtime points bound back to source channel ids, with per-apartment point templates for the two submeter profiles.
- 15 gateway profiles covering 21 format cases, plus the HMR handheld-event publisher.

Regenerate the register and packet with `python generators/northlake_packet.py` (check first with `--check`), then update the Data Collection workbook with `python generators/update_workbook.py`.

The inventory deliberately includes both ordinary and awkward shapes:

Provider	Meter shape	Channels
Valley Electric District	Monthly electric meters plus interval demand meters, and unmetered area-lighting services	kWh billing usage, kW billing demand, 15-minute kWh, 15-minute kW, flat fixture charge with no usage
Sierra Gas Utility	Monthly gas meters	therms, with ccf as delivered volume metadata
River City Utilities	Water meters on bundled accounts, irrigation meters (three on the Campus Grounds account) and flat-rate fire service lines	water kgal, derived sewer kgal, flat stormwater ERU where applicable, irrigation kgal, flat fire-service charge
Northlake Thermal Plant	Chilled-water BTU meters at two served buildings plus steam meters at six, all under university buildings	hourly ton-hours, hourly tons, monthly steam klb
Helios Onsite Solar	PV production/export meters	generated kWh, exported kWh, generated kW
Owned meters (no provider)	Apartment and house submeters, sewer-deduct submeters, Central Plant production meters, lab sensors, the Library historian, the chiller controller, the smoke-test source	hourly submeter kWh, route register and usage kgal, manual deduct readings, hourly ton-hours, klb and kgal production, 5-minute chiller kW and status, sensor and historian interval values
Reference channels	KSAC and KSMF weather stations	temperature, humidity, wind, forecast temperature

Gateway Proof Points

The gateway bindings in the scenario manifest are written against the current EEMSuite gateway component contract in `EEMSuite/src/Tasks/EEMSuite.Tasks.Gateways`:

Gateway profile	EEMSuite component	Source provider	Points proved
gw_acquisuite_student_elect ric	AcquiSuiteGateway (ACQUISUITE)	Valley Electric District	Student Center and Lakeview Residence Hall 15-minute kWh and kW, using Serial Number, DeviceID, and Row Number bindings
gw_noaa_ksac_observations	NOAAGateway (NOAA)	KSAC weather station	NOAA station observations with WeatherType, StationID, and WeatherTypeName mappings
gw_aeris_ksmf_weather	AerisWeatherGateway (AERISWEATHER)	KSMF weather station	Aeris observations, forecasts, state-file behavior, and weather metric names
gw_mv90_valley_mdef	MV90Gateway (MDEF)	Valley Electric District	Science Center, Recreation and Aquatics Center and Central Plant 15-minute kWh and kW, using Recorder ID and Channel bindings
gw_mv90_helios_mv9	MV90Gateway (MV9)	Helios Onsite Solar	Solar generation/export interval points, including positive-generation sign convention
gw_bacnet_thermal_plant	BACnetGateway (BACNET)	Northlake Thermal Plant	Chilled-water ton-hours and tons at Science Center and Recreation and Aquatics Center, plus Central Plant chilled-water supply, steam production and condensate return, from BACnet present values
gw_spinwave_science_lab	SpinwaveGateway (SPINWAVE)	Science Center sensors	Spinwave folder, device address, and channel-number file mapping
gw_fig_multi_format	FIGGateway (FIG)	River City Utilities alias data	FIG, SIEMANSREPORT, MEDIATOR, EATON, XML, FIXEDNETWORKS, and CMEP format variants
gw_modbus_thermal_plant	ModbusGateway (MODBUS)	Northlake Thermal Plant controller	Modbus TCP node settings and floating-point register decoding
gw_odbc_library_historian	ODBCGateway (ODBC)	Synthetic SQL historian	Profile/query-key lookup, token substitution, timestamp cursor state, and backfill behavior
gw_fake_smoke	FakeGateway (FAKE)	Synthetic generated channel	Deterministic generated interval data, generation modes, and state handling
gw_neptune_river_city_water	NeptuneGateway (NEPTUNE)	River City Utilities	Neptune handheld water upload parsing, usage point, and register-reading point mapping
gw_mvrs_sierra_gas	MVRSGateway (MVRS)	Sierra Gas Utility	MVRS handheld gas upload parsing, usage/register points, read-code events, and HMR event file output

Gateway profile	EEMSuite component	Source provider	Points proved
gw_fig_fixed_network_submeters	FIGGateway (FIXEDNETWORKS)	Fixed-network electric submeters	Hourly kWh for every Cedar Row apartment and house electric submeter, the Lakeview Residence Hall laundry and the Parking Structure EV chargers, using per-apartment point templates
gw_neptune_cedar_row_submeters	NeptuneGateway (NEPTUNE)	Cedar Row apartment water submeter route	Monthly register and usage points for every Cedar Row apartment and house water submeter
hmr_events_publish	HmrEventsPublish (HMR_EVENTS)	MVRS/Neptune outputs	HMR event JSONL publishing, idempotency scope, queue naming, and gateway identity metadata

The source model treats channels as the business facts and gateway points as collector bindings. Generated AcquiSuite logs, MDEF/MV9 files, BACnet runtime JSON, handheld upload files, HMR event payloads, BIF files, and assertions should all trace back to the same source channel ids.

Required Patterns

V1 should include normal data and deliberately odd but realistic events:

- Seasonal electric usage that correlates with weather.
- Lab building baseload higher than classroom buildings.
- Student Center demand spikes during event periods.
- Apartment water usage that scales with occupancy.
- One meter replacement.
- One estimated bill followed by a corrected bill.
- One duplicate import attempt.
- One missing interval block that is later backfilled.

Success Criteria

The V1 customer is useful when it can:

- Load into a dev environment without production data.
- Produce repeatable rollups by customer, site, building, provider, account, and meter.
- Exercise every gateway component in `EEMSuite.Tasks.Gateways`.
- Exercise at least one backfill path.
- Exercise at least one external integration export path.
- Produce known expected totals for reports and validation checks.